



D π D Maths

TMUA

Mock Set 1

Paper 1

20 Multiple Choice Questions

Time allowed: 75 minutes

Calculators are not permitted

dpidmaths.co.uk

YouTube: @DPiDMaths

Question 1

Let k be a real number. The equation $|x^2 + 3kx| = 2k|x - 3|$ has n solutions for a given k . Find the sum of all values of n in $\{0, 1, 2, 3, 4, 5\}$ that are possible.

- A) 6
- B) 9
- C) 10
- D) 12
- E) 14
- F) 15

$$K < 0:$$

LHS non negative so for there to be a solution $LHS = RHS = 0$

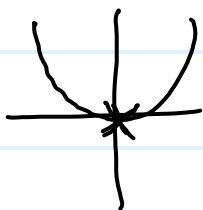
$$2K|x-3| = 0, \quad x=3$$

$$|9+9K| = 0, \quad K = -1$$

$$K = -1, \quad n = 1 \quad \text{otherwise } K < 0, K \neq -1, n = 0$$

$$K = 0:$$

$$|x^2| = 0$$



$$n = 1$$

$$K > 0:$$

$$2K|x-3| = |2K(x-3)|, \quad |x^2+3Kx| = |2K(x-3)|$$

$$x^2 + 3Kx = \pm 2K(x-3)$$

$$Q_1: x^2 + 3Kx = 2Kx - 6K$$

$$x^2 + Kx + 6 = 0 \quad \Delta_1: K^2 - 24K, \quad K > 0$$

$0 < K < 24 \rightarrow 0 \text{ roots}, \quad K = 24 \rightarrow 1 \text{ root}, \quad K > 24 \rightarrow 2 \text{ roots}$

$$Q_2: x^2 + 3Kx = 2Kx + 6K$$

$$x^2 + 5Kx - 6K = 0 \quad \Delta_2: 25K^2 + 24K$$

$K > 0 \rightarrow 2 \text{ roots always}$

both Quadratics considered:

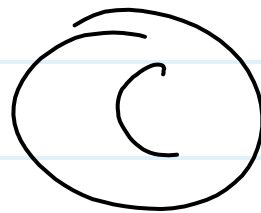
$$0 < k < 24 \rightarrow 2 \text{ solutions}$$

$$k = 24 \rightarrow 3 \text{ solutions}$$

$$k > 24 \rightarrow 4 \text{ solutions}$$

Overall $n = 0, 1, 2, 3, 4$, $n \neq 5$

$$1+2+3+4 = 10$$



Question 2

Find the complete set of x for which $\frac{1}{\sqrt{6 - \log_3(x^4)}}$ is defined.

- A) $-27 < x < 27, x \neq 0$
- B) $-3\sqrt{3} < x < 3\sqrt{3}, x \neq 0$
- C) $-9 < x < 9, x \neq 0$
- D) $0 < |x| < 27$
- E) $-3 < x < 3, x \neq 0$
- F) $|x| \geq 27$
- G) $-27 \leq x \leq 27$
- H) $-3\sqrt{3} \leq x \leq 3\sqrt{3}$

$$\frac{1}{\sqrt{6 - \log_3(x^4)}}$$

restrictions $\frac{1}{y} \in \mathbb{R}, y \neq 0 \rightarrow \sqrt{6 - \log_3(x^4)} \neq 0$
 $\sqrt{p} \in \mathbb{R}, p \geq 0 \quad 6 - \log_3(x^4) \geq 0$
 $\log_3(u) \in \mathbb{R}, u \geq 0 \quad x^4 \geq 0$

combining gives:

$$6 - \log_3(x^4) > 0$$

$$\log_3(x^4) < 6$$

$$x^4 < 729$$

$$|x| < (729)^{1/4} \rightarrow |x| < (3^6)^{1/4}$$

$$|x| < 3^{3/2} \quad |x| < 3\sqrt{3}$$

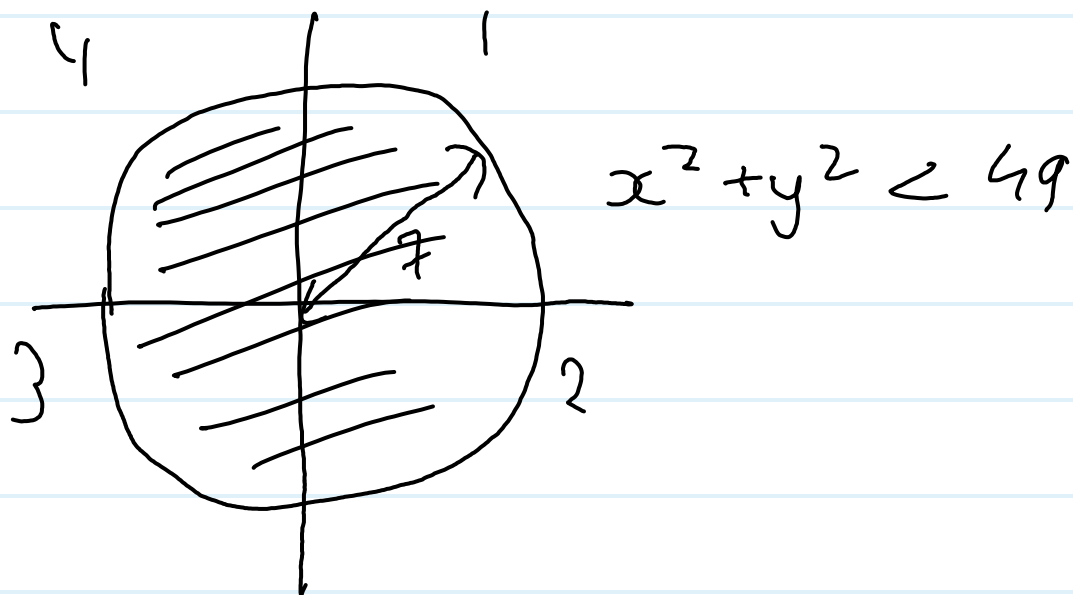
$$-3\sqrt{3} < x < 3\sqrt{3}$$

(B)

Question 3

A circle of radius 7 is drawn centred at the origin. How many squares of side length 1 and integer coordinate vertices intersect the interior of this circle?

- A) 156
- B) 164
- C) 168
- D) 172
- E) 176
- F) 180
- G) 184
- H) 196



- consider a square in quadrant 1 with a bottom left corner (a, b) where $a, b \in \mathbb{Z}^+$
- Square intersects interior in 1st quadrant if (a, b) in circle i.e. $a^2 + b^2 < 49$

when $a = 0$ $b^2 < 49$, $b = 0, 1, 2 \dots 6$ total 7

when $a = 1$ $b^2 < 48$, $b = 0, 1, 2 \dots 6$ total 7

when $a = 2$ $b^2 < 45$, $b = 0, 1, 2 \dots 6$ total 7

$a = 3$ $b^2 < 40$, $b = 0, 1, 2, \dots 6$ total 7

$a = 4$ $b^2 < 36$, $b = 0, 1, 2, \dots 5$ total 6

$a = 5$ $b^2 < 24$, $b = 0, 1, 2, 3, 4$ total 5

$a = 6$ $b^2 < 13$, $b = 0, 1, 2, 3$ total 4

Hence in Q, there is $7 \times 4 + 6 + 5 + 4 = 43$

Symmetry in 4 quadrants $= 4 \times 43 = 172$ (D)

Question 4

Let x be a real number. Find the maximum value of 3^{6x-2x^2-5} .

A) $\frac{\sqrt{3}}{3}$

B) $\sqrt{3}$

C) 3

D) $\frac{1}{3}$

E) $\frac{1}{9}$

F) $\frac{\sqrt{2}}{2}$

G) 1

H) $3\sqrt{3}$

$$\text{maximise } 3^{6x - 2x^2 - 5}$$

$$\begin{aligned} 6x - 2x^2 - 5 &= -2 \left(x^2 - 3x + \frac{5}{2} \right) \\ &= -2 \left(x - \frac{3}{2} \right)^2 - \frac{1}{2} \quad \text{complete square} \end{aligned}$$

$$\text{max value at } x = \frac{3}{2}$$

$$3^{-1/2} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3} \quad \textcircled{A}$$

Question 5

Define $P(x) = (x-1^2)(x-2^2) \cdots (x-100^2)$. How many integers n are there such that $P(n) \leq 0$?

- A) 4900
- B) 4950
- C) 5000
- D) 5050
- E) 5100

$$P(x) = (x-1^2)(x-2^2)(x-3^2) \dots (x-100^2)$$

$$P(x) = 0 \text{ at } x = 1^2, 2^2, 3^2, 4^2 \dots 100^2$$

$$\deg(P(x)) = 100 \text{ so}$$

$$P(x) > 0, x > 100^2$$

for $P(x) < 0$ odd number of negative values:

$$\text{when } 1^2 \leq x \leq 2^2, 3^2 \leq x \leq 4^2 \dots$$

$$99^2 \leq x \leq 100^2$$

can be expressed as:

$$(2k-1)^2 \leq x \leq (2k)^2, k = 1, 2, 3 \dots 50$$

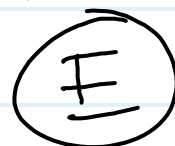
valid number of values is given by

$$(2k)^2 - (2k-1)^2 + 1$$

$$4 \sum_{k=1}^{50} k = 4 \left(\frac{50 \times 51}{2} \right)$$

$$= 4k^2 - 4k^2 + 4k - 1 + 1 = 5100$$

$$= 41k$$



Question 6

Find the constant term of $\left(|x| + \frac{1}{|x|} - 3\right)^3$.

- A) 20
- B) -30
- C) -45
- D) -27
- E) 45
- F) 18
- G) -18
- H) 27

constant

$$\Rightarrow (-3)^3 = -27$$

$$+ \frac{3!}{1!1!1!1!} \times -3 \times \frac{1}{|x|} \times |x|$$

$$= 3! \times -3 = -18$$

$$-27 - 18 = -45 \quad \textcircled{C}$$

Question 7

Real numbers a, b, c satisfy $a + b + c = 0$ with $0 < a < c$. Find the number of solutions to $c \sin^2 \theta + b \sin \theta + a = 0$ in $0 < \theta < \pi$.

- A) 3
- B) 1
- C) 2
- D) 4
- E) 0
- F) 5
- G) 6
- H) Infinite

$$a + b + c = 0, \quad 0 < a < c$$

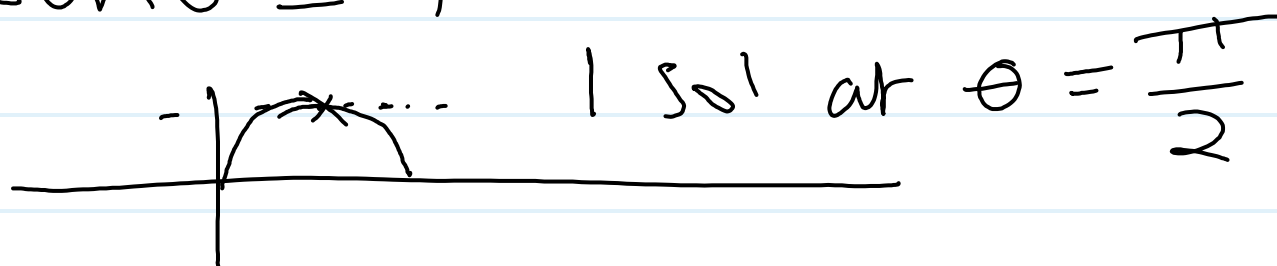
$$b = -(a + c) \quad 0 < \theta < \pi$$

$$c \sin^2 \theta + b \sin \theta + a = 0$$

$$c \sin^2 \theta - (a + c) \sin \theta + a = 0$$

$$(c \sin \theta - a)(\sin \theta - 1) = 0$$

$$\sin \theta = 1$$



$$\sin \theta = \frac{a}{c} \quad \text{as } 0 < a < c$$

$$0 < \frac{a}{c} < 1 \rightarrow$$

~~1 sol~~ 2 solutions

1 + 2 = total 3 solutions

(A)

Question 8

The ellipse is $\frac{x^2}{9} + y^2 = 1$. Given point $M(2, 0)$ and point P on the ellipse, find the minimum value of $|PM|$.

- A) $\frac{\sqrt{2}}{3}$
- B) $\frac{\sqrt{3}}{3}$
- C) $\frac{\sqrt{6}}{3}$
- D) $\frac{2\sqrt{2}}{3}$
- E) $\frac{\sqrt{2}}{2}$
- F) 1
- G) $\frac{1}{3}$
- H) $\frac{1}{2}$

Let P have coords (m, n)

$$(|PM|)^2 = (m-2)^2 + n^2$$

$$\frac{m^2}{9} + n^2 = 1$$

$$n^2 = 1 - \frac{m^2}{9}$$

$$(m-2)^2 + 1 - \frac{m^2}{9} = (|PM|)^2$$

$$m^2 - 4m + 4 + 1 - \frac{m^2}{9} = (|PM|)^2$$

$$= \frac{8}{9}m^2 - 4m + 5$$

$$= \frac{8}{9} \left[m^2 - \frac{9}{2}m \right] + 5$$

$$= \frac{8}{9} \left(m - \frac{9}{4} \right)^2 + \frac{1}{2}$$

$$(|PM|)_{\min}^2 = \frac{1}{2} \quad |PM|_{\min} = \frac{\sqrt{2}}{2}$$



Question 9

Given $f(x) = |x^2 - 10x + 13|$ and $g(x) = 6x + \frac{6}{x}$. With constant $m \in \mathbb{R}$, let $\epsilon = f(x) - g(m)$, where $\epsilon^2 + 12\epsilon = 0$. Find the maximum number of solutions of x .

- A) 1
- B) 2
- C) 3
- D) 4
- E) 5
- F) 6
- G) 7
- H) 8

$$\xi^2 + 12\xi = 0$$

$$\xi(\xi + 12) = 0 \quad \xi = 0 \quad \text{or} \quad \xi = -12$$

$$\text{Let } g(m) = y$$

$$y = 6m + \frac{6}{m}, \quad ym = 6m^2 + 6$$

$$6m^2 - ym + 6 = 0 \rightarrow$$

$$y, m \in \mathbb{R} \text{ hence } \Delta \geq 0$$

$$\Delta: \quad y^2 - 4(6 \times 6) \geq 0$$

$$y^2 - 144 \geq 0$$

$$|y| \geq 12$$

$$f(x) = |x^2 - 10x + 13|$$

$$\text{if } |x^2 - 10x + 13| = k$$

$$k \geq 0$$

$$|x^2 - 10x + 13| = k$$

$$x^2 - 10x + 13 = \pm k$$

$$Q_1: x^2 - 10x + 13 + k = 0$$

$$Q_2: x^2 - 10x + 13 - k = 0$$

$$\Delta_1: 100 - 4(13 + k) = 4(12 + k)$$

$$\Delta_2: 100 - 4(13 - k) = 4(12 - k)$$

for all $k \geq 0$ $\Delta_1 > 0$ hence
 Q_1 has 2 real solutions for $k \geq 0$

$$\Delta_2:$$

$$0 < k < 12 \rightarrow 2 \text{ solutions}$$

$$k = 12 \rightarrow 1 \text{ repeated sol}$$

$$k > 12 \rightarrow 0 \text{ solutions}$$

considering both Quadratics:

$$k = 0 \rightarrow 2 \text{ solutions}$$

$$0 < k < 12 \rightarrow 4 \text{ solutions}$$

$$k = 12 \rightarrow 3 \text{ solutions}$$

$$k > 12 \rightarrow 2 \text{ solutions}$$

$$f(x) = y$$

$$y \geq 0$$

$$f(x) = y - 12$$

$$y - 12 \geq 0$$

The 'limiting' inequality is
 $y \geq 12$

inequalities:

$$f(x) = y$$

sols

$$f(x) = y - 12$$

sols

$$y = 12$$

$$12 < y < 24$$

$$y = 24$$

$$y > 24$$

3	2
2	4
2	3
2	2

↑

F

maximum
 total $2 + 4 = 6$

Question 10

Let $a_n = \frac{n \cdot 2^{n+1}}{(n+1)(n+2)}$, where n is a positive integer. Find $a_1 + a_2 + a_3 + \cdots + a_{254}$.

A) $\frac{2^{256}}{256}$

B) $\frac{2^{256}}{256} - 2$

C) -2

D) $\frac{2^{256}}{256} + 2$

E) 2

F) $\frac{2^{255}}{255} - 2$

G) $\frac{2^{256}}{255} - 2$

H) $\frac{2^{254}}{256} - 2$

$$\frac{n \cdot 2^{n+1}}{(n+1)(n+2)} = a_n$$

$$\frac{n}{(n+1)(n+2)} \equiv \frac{A}{n+1} + \frac{B}{n+2}$$

$$n \equiv A(n+2) + B(n+1)$$

$$A + B = 1 \quad 2A + B = 0$$

$$A = -1, \quad B = 2$$

$$\frac{n}{(n+1)(n+2)} = \frac{-1}{n+1} + \frac{2}{n+2}$$

$$\sum_{n=1}^{254} a_n = \sum_{n=1}^{254} \left(\frac{2^{n+2}}{n+2} - \frac{2^{n+1}}{n+1} \right)$$

$n=1:$

$$\frac{2^3}{3} - \frac{2^2}{2}$$

$n=2:$

$$\frac{2^4}{4} - \frac{2^3}{3}$$

$n=3:$

$$\frac{2^5}{5} - \frac{2^4}{4}$$

$\vdots$
 $\vdots$
 $\vdots$
 $\vdots$

$n=253:$

$$\frac{2^{255}}{255} - \frac{2^{254}}{254}$$

$n=254:$

$$\frac{2^{256}}{256} - \frac{2^{255}}{255}$$

$$\frac{2^{256}}{256} - \frac{2^2}{2}$$

$$= \frac{2^{256}}{256} - 2$$

(13)

Question 11

A curve $y = x^2 - 4x$ is rotated 180° about the origin, then translated by $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$. Find the equation of the new curve.

A) $y = 7 - x^2$

B) $y = 1 - x^2$

C) $y = -1 - x^2$

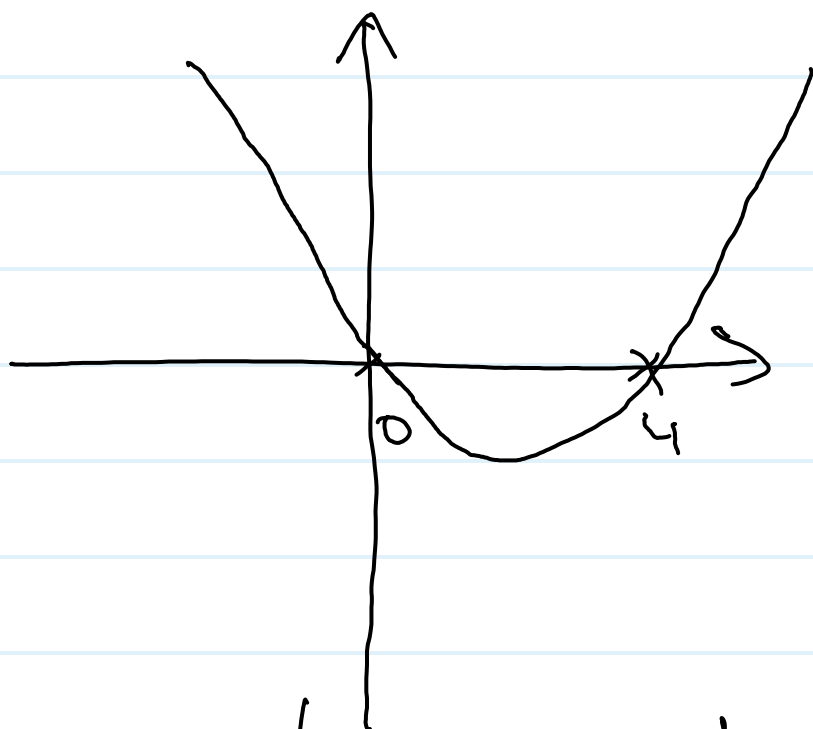
D) $y = 3 - x^2$

E) $y = 7 - (x - 2)^2$

F) $y = 7 + x^2$

G) $y = -7 - x^2$

H) $y = 4 - x^2$

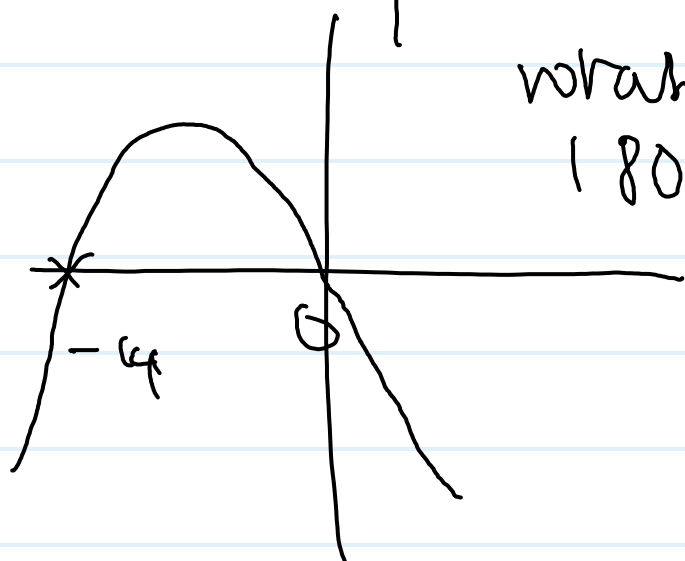


$$x^2 - 4x$$

$$x(x - 4)$$

$$(x - 2)^2 - 4$$

min: $(2, -4)$



rotated
 180°

$$(x, y) \rightarrow (-x, -y)$$

new maximum point
is $(-2, 4)$

$$\begin{pmatrix} -2 \\ 4 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 7 \end{pmatrix} \rightarrow \text{final maximum point}$$

$$-(x - 0)^2 + 7$$

$$= -x^2 + 7$$

$$= 7 - x^2$$

Ⓐ

Question 12

Suppose x and y are positive reals such that $x - y^2 = 1$ and $x^2 + y^4 = 41$. Find x .

A) 4

B) 5

C) $\sqrt{41}$

D) 6

E) $\frac{-1 + \sqrt{17}}{2}$

F) 3

G) 20

H) 9

$$x - y^2 = 1 \quad x^2 + y^4 = 41$$

$$y^2 = x - 1$$

$$y^4 = x^2 - 2x + 1$$

$$x^2 + x^2 - 2x + 1 = 41$$

$$2x^2 - 2x - 40 = 0$$

$$x^2 - x - 20 = 0$$

$$(x - 5)(x + 4) = 0$$

$$x = 5 \text{ or } -4$$

$$y^2 \geq 0, \quad y \in \mathbb{R}$$

$$y^2 = x - 1$$

$$x - 1 \geq 0 \quad x \geq 1$$

$$x = 5$$

(B)

Question 13

A rectangular sheet of paper is folded along its diagonal, making a pentagon that has an area of $\frac{11}{16}$ of the area of the original rectangle. Find the ratio of the longer side of the rectangle to the shorter side.

A) $\frac{11}{16}$

B) $\sqrt{3}$

C) 2

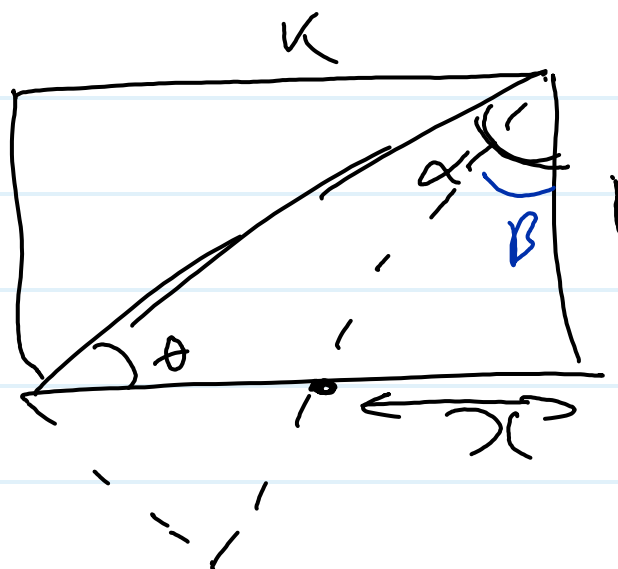
D) $\frac{5}{2}$

E) $\sqrt{5}$

F) $\frac{3}{2}$

G) 4

H) $\sqrt{2}$



$$A_{og} = k$$

$$A_{fold} = \frac{1}{2}(k-x) + x$$

$$\tan \theta = \frac{1}{k} \quad \tan \alpha = k \quad \tan \beta = x$$

$$\alpha = \theta + \beta$$

$$\tan(\theta + \beta) = \frac{\tan \theta + \tan \beta}{1 - \tan \theta \tan \beta}$$

$$k = \frac{\frac{1}{k} + x}{1 - \frac{x}{k}}$$

$$K \left(1 - \frac{x}{K} \right) = \frac{1}{K} + x$$

$$K - x = \frac{1}{K} + x$$

$$2x = K - \frac{1}{K} \quad x = \frac{1}{2} \left(K - \frac{1}{K} \right)$$

$$\frac{11}{16} A_{\text{og}} = A_{\text{old}}$$

$$\frac{11}{16} K = \frac{1}{2} (K - x) + x$$

$$= \frac{1}{2} \left(K - \frac{1}{2} K + \frac{1}{2K} \right) + \frac{1}{2} \left(K - \frac{1}{K} \right)$$

$$= \frac{1}{4} K + \frac{1}{4K} + \frac{1}{2} K - \frac{1}{2K}$$

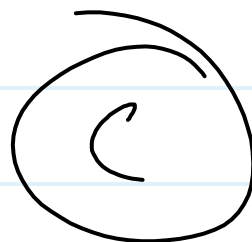
$$= \frac{3}{4} K - \frac{1}{4K}$$

$$\frac{11}{16}K = \frac{3}{4}K - \frac{1}{4K}$$

$$\frac{1}{16}K - \frac{1}{4K} = 0$$

$$K^2 - 4 = 0$$

$$K^2 = 4 \quad K = 2$$



Question 14

Tom needs to pay £100 for his dinner. He has an unlimited supply of £2 coins and £5 and £10 notes. In how many ways can he pay?

- A) 60
- B) 63
- C) 66
- D) 55
- E) 121
- F) 36
- G) 72
- H) 100

$$100 = 2x + 5y + 10z$$

$$x, y, z \in \mathbb{Z}^+$$

as $5y$ and $10z$ multiple of 5, $2x$ is also multiple of 5 so x is a multiple of 5 as well

hence let $x = 5k$

$$10k + 5y + 10z = 100$$

$$\underbrace{2k}_{\text{even}} + y + \underbrace{2z}_{\text{even}} = \underbrace{20}_{\text{even}}$$

hence y must be even, let $y = 2j$

$$2k + 2j + 2z = 20$$

$$k + j + z = 10$$

num of solutions are same as the og equation.

when k is 0:

$$j + z = 10$$

$$0 + 10, 1 + 9 \dots 10 + 0 \rightarrow 11 \text{ sols}$$

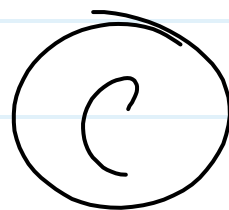
k is 1:

$$j + z = 9$$

$$0 + 9, 1 + 8 \dots 9 + 0 \rightarrow 10 \text{ sols}$$

$$11 + 10 + 9 + 8 \dots 1$$

$$= \frac{11 \times 12}{2} = 66$$



Question 15

What are the last 6 digits of $11 \times 101 \times 1001 \times 10001 \times 11$?

- A) 344321
- B) 443221
- C) 222111
- D) 443321
- E) 344421
- F) 421443
- G) 434221
- H) 442321

we only care about the 6
least significant digits:

$$11 \times 101 = 1100 + 11 = 1111$$

$$1111 \times 1001 = 1111000 + 1111 \\ = \underline{1112111}$$

$$112111 \times 10001 \\ = 1121110000 + 112111 \\ = 1121\underline{222111}$$

$$222111 \times 11 = 2221110 + 222111 \\ = \underline{2443221}$$

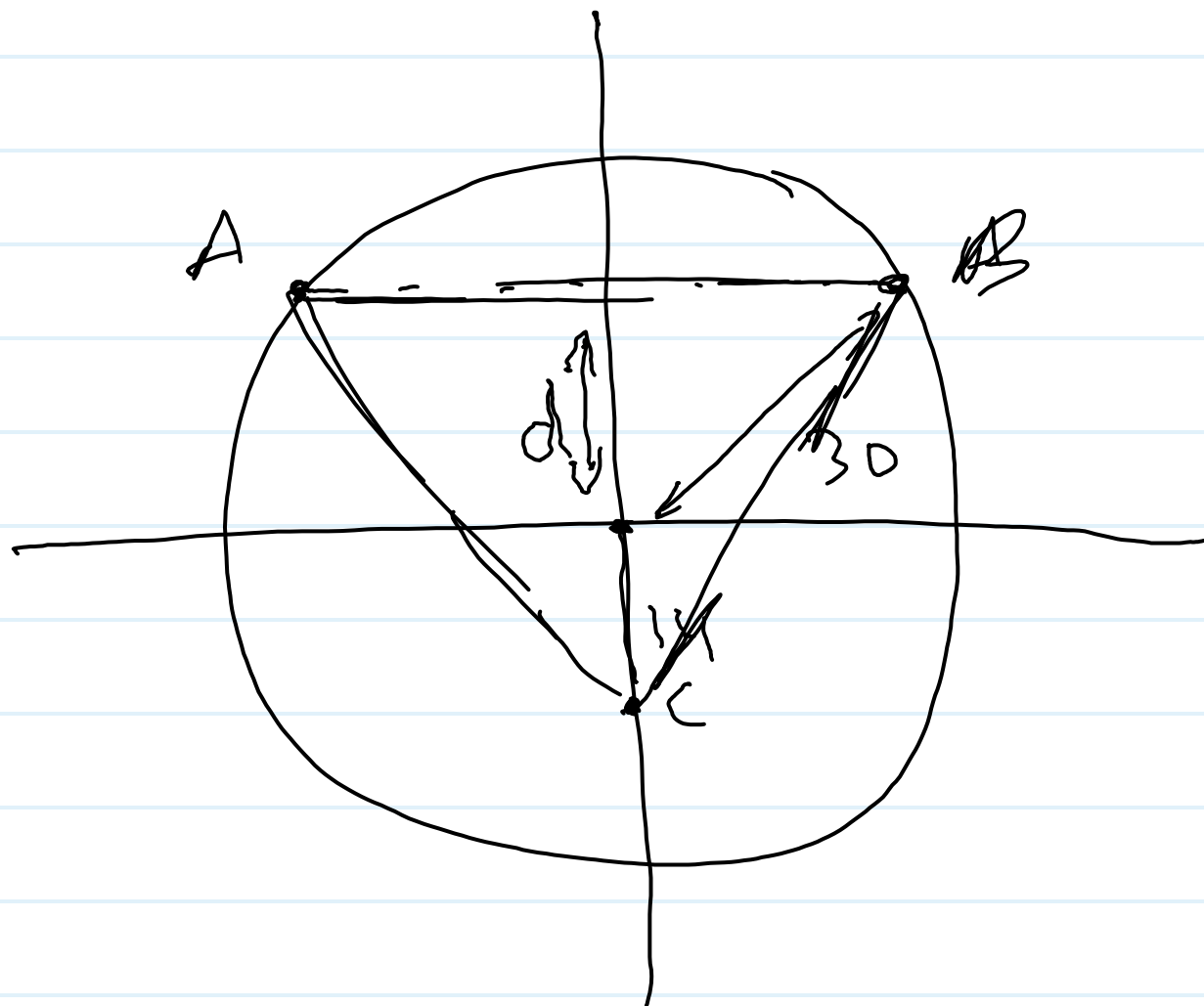
4,43221

(B)

Question 16

Let O be the point $(0,0)$. Let A, B, C be three points in the plane such that $OA = 30$, $OB = 30$, and $OC = 14$, chosen so that the area of triangle ABC is maximal. What is the length of the shortest side of ABC ?

- A) 44
- B) 40
- C) 48
- D) 34
- E) 30
- F) 42
- G) 38
- H) 36



to maximise area we want the height to be maximised away from the base so put it directly behind O

$$(or |AB| = 1)$$

$$\text{Pythag: } \left(\frac{1}{2}\right)^2 + d^2 = 30^2$$

$$\frac{D^2}{4} + d^2 = 900$$

$$D = 2\sqrt{900 - d^2}$$

★ hence the height of triangle is $d + 14$

$$A(d) = \frac{1}{2} \times 2\sqrt{900 - d^2} (d + 14)$$

$$= (d + 14)\sqrt{900 - d^2}$$

$$\text{or } g(d) = (A(d))^2$$

$$g(d) = (d^2 + 28d + 196)(900 - d^2)$$

$$= -d^4 - 28d^3 + 704d^2 + 25200d + C$$

$$g'(d) = 0$$

$$= d^3 + 21d^2 - 352d - 6300 = 0$$

$$= (d - 18)(d + 14)(d + 25) = 0$$

$$d = 18, -14, -25$$

$$d > 0, \quad d = 18$$

$$D = 2\sqrt{900 - 324} = 2\sqrt{576}$$

$$= 48$$

$$AC = \sqrt{24^2 + 32^2} \rightarrow 40$$

8 x 3, 4, 5 triangle

(B)

Question 17

Alice and Bob repeatedly flip a fair coin. Alice wins if 1 head followed by 3 tails appears. Bob wins if 5 heads in a row appears. They flip until someone wins. What is the probability that Alice wins?

A) $\frac{1}{2}$

B) $\frac{15}{19}$

C) $\frac{7}{11}$

D) $\frac{3}{5}$

E) $\frac{15}{16}$

F) $\frac{8}{23}$

G) $\frac{16}{31}$

H) $\frac{31}{47}$

Alice wins: HTTTT

Bob wins: HHHHH

if the start is just tails we can disregard them until the first head.

Let p = probability Alice wins after the first H appears.

Suppose we are at HTT

↳ if next is T Alice wins

↳ if next is H it resets

$$\begin{aligned} \text{hence } P(\text{HTT}) &= \frac{1}{2} + \frac{1}{2}p \\ &= \frac{1+p}{2} \end{aligned}$$

suppose now we are at HT

↳ if next is HT $\boxed{T}$ → where it is $\frac{1+p}{2}$

↳ if next is HT $\boxed{H}$ → resets

Hence probability Alice wins with HT

$$P(HT) = \frac{1}{2} \left(\frac{1+p}{2} \right) + \frac{1}{2} p$$

$$= \frac{1+3p}{4} = \frac{1}{2}$$

consider sum of HHHH

P Alice wins from here is $P(HHHH)$

$$= \frac{1}{2}$$

Same logic:

$$P(HHHH) = \frac{1}{2} \frac{1}{2} + \frac{1}{2} \left(\frac{1}{2} \right) = \frac{3}{4}$$

$$P(HH) = \frac{1}{2}q + \frac{1}{2}\left(\frac{3}{4}q\right) = \frac{7}{8}q$$

$$P(H) = \frac{1}{2}q + \frac{1}{2}\left(\frac{7}{8}q\right) = \frac{15}{16}q$$

$$p = P(H) = \frac{15}{16}q$$

$$p = \frac{15}{16} \left(\frac{1+3p}{4} \right)$$

$$64p = 15 + 45p$$

$$19p = 15$$

$$p = 15/19$$

(B)

Question 18

Let $p(x)$ and $q(x)$ be two cubic polynomials such that $p(0) = -6$, $q(0) = 10$, and $p(q(x)) = q(p(x))$ for all real numbers x . Find the ordered pair $(p(2), q(4))$.

- A) $(2, -6)$
- B) $(-6, 2)$
- C) $(2, 6)$
- D) $(-6, 10)$
- E) $(6, -6)$
- F) $(2, -10)$
- G) $(-2, 6)$
- H) $(10, -6)$

$$\text{Let } p(x) = ax^3 + bx^2 + (x - 6)$$

$$\text{Let } q(x) = Ax^3 + Bx^2 + (x + 10)$$

$$p(q(x)) = q(p(x))$$

$$p(q(x)) = aA^3$$

$$a(Ax^3 + Bx^2 + Cx + 10)^3$$

$$\rightarrow aA^3x^9 + 3aA^2Bx^8 +$$

$$\left(\frac{3!}{2!} (A^2x^6) + \frac{3!}{2!} (B^2x^4Ax^3) \right) a$$

$$= aA^3x^9 + 3aA^2Bx^8 + (3aA^2C + 3aAB^2)x^7 + \dots$$

$$q(p(x))$$

$$A(ax^3 + bx^2 + (x - b))$$

$$= Aa^3x^9 + 3Aa^2bx^8 + (3Aa^2c + 3Aab^2)x^7 + \dots$$

comp coeffs:

$$aA^3 = Aa^3 \quad A^2 = a^2$$

$$A = \pm a \quad A = ka \quad k \in \{1, -1\}$$

$$3a\cancel{A^2}B = 3A\cancel{a^2}b$$

$$3aB = 3Ab$$

$$B = kb$$

$$\cancel{3}a\cancel{A}^2\cancel{C} + \cancel{3}a\cancel{A}\cancel{B}^2 = \cancel{3}\cancel{A}\cancel{a}^2\cancel{C} + \cancel{3}\cancel{A}a\cancel{B}^2$$

$$AC + \cancel{B}^2 = ac + \cancel{b}^2$$

$$AC = ac$$

$$C = kC$$

test

is $k=1$?

if $k=1$

$$q(x) = p(x) + 16$$

$$p(p(x) + 16) = p(p(x)) + 16$$

for all real x

if $x \in \mathbb{R}$ then $p(x) \in \mathbb{R}$
 or $y = p(x)$

$$p(y+16) = p(y) + 16$$

$$p(y+16) - p(y) = 16$$

This is not possible because for a cubic p this above is a quadratic which cannot be 16 for all y ; hence $k = -1$.

$$k = -1:$$

$$q(x) = 4 - p(x)$$

$$p(4 - p(x)) = 4 - p(p(x))$$

$$\text{or } y = p(x)$$

$$x \in \mathbb{R}, y \in \mathbb{R}$$

$$p(4 - y) = 4 - p(y)$$

$$P(2) = 4 - P(2)$$

$$P(2) = 2$$

$$q(4) = 4 - P(4)$$

$$4 - P(4) = P(0)$$

$$q(4) = P(0) = -6$$

$$(P(2), q(4)) = (2, -6)$$

A

Question 19

An ant starts at the origin, facing in the positive x -direction. Each second, it moves 1 unit forward, then turns counterclockwise by $\sin^{-1}\left(\frac{5}{13}\right)$ degrees. What is the least upper bound on the distance between the ant and the origin?

A) $\sqrt{13}$

B) $\sqrt{26}$

C) $\frac{13}{5}$

D) 5

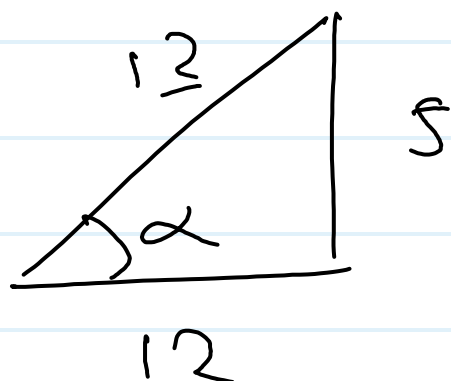
E) $\frac{13}{2}$

F) $\sqrt{10}$

G) $2\sqrt{13}$

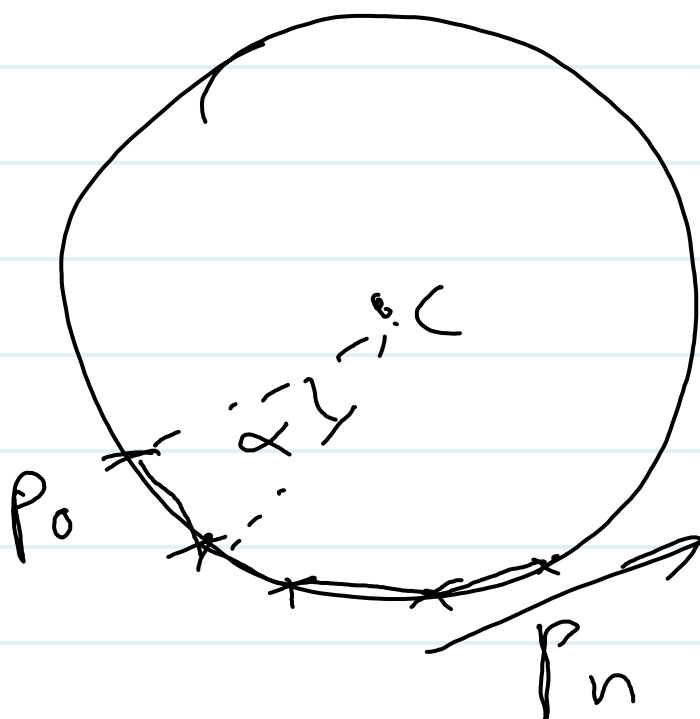
H) $\frac{26}{5}$

$$\text{Let } \alpha = \sin^{-1} (5/13)$$



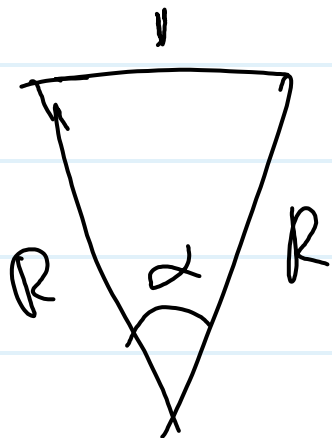
$$\cos \alpha = \frac{12}{13}$$

$$\sin \alpha = 5/13$$



Ant moves
through a
circle
each time
turning α
in the circle

Let circle have radius R



cosine rule:

$$1 = R^2 + R^2 - 2R^2 \cos \alpha$$

$$1 = 2R^2 - 2R^2 \left(\frac{12}{13} \right)$$

$$1 = 2R^2 \left(1 - \frac{12}{13} \right)$$

$$\frac{13}{2} = R^2 \quad R = \sqrt{\frac{13}{2}}$$

The furthest the ant can be from the origin is the diameter of the circle $= 2R$

$$2R = 2\sqrt{\frac{13}{2}} = \sqrt{26}$$

(B)

Question 20

Ava and Ben play a game on a 12×12 board. For some positive integer k , Ava chooses k squares and writes an A in each of them. Ben then chooses $k + 3$ squares and writes a B in each. After Ben is done, Ava wins if there is a sequence of letters forming “ BAA ” or “ AAB ” when read either horizontally (left-to-right) or vertically (top-to-bottom), and Ben wins otherwise. Determine the smallest value of k for which Ava has a winning strategy.

- A) 24
- B) 34
- C) 35
- D) 36
- E) 37
- F) 40
- G) 48
- H) 18

Let an empty square be called 'rotten' if when Ben puts a B in it it creates a 'BAA' or 'AAB'

A $\rightarrow$ Alice

R $\rightarrow$ rotten

S $\rightarrow$ safe

B $\rightarrow$ Ben

Consider a 4×4 grid.

S	R	R	S
R	A	A	R
R	A	A	R
S	R	R	S

every R becomes BAA or AAB if Ben puts a B on it.

4 A's lead to 8 rotten in a 4×4

a 12×12 is a 3×3 array of (4×4) 's.

if you repeat the pattern in
each 4×4 block you use
 $4 \times 9 = 36$ As

This creates $8 \times 9 = 72$ Rs
and only 36 safe blocks

for Ben to win he must
place $k+3 = 39$
but as $39 > 36$ he
cannot win so

for 36 Ava has a winning
strategy.

The minimum is 36 because:

consider looking at just horizontally,
in any row a rotten square must
be next to at least 2 consecutive
A's.

→ a run of 2 A's uses at least
2 A's and creates at most 2
rotten squares

hence num of horizontal rotten $\leq k$

wlog for vertical

num of vertical rotten $\leq k$

so total at most $2k$ rotten squares

So number of safe empty squares is at least

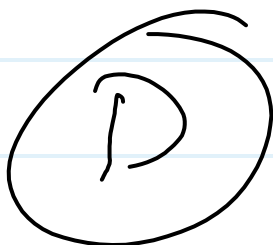
$$144 - 2K - K = 144 - 3K$$

$$144 - 3K \geq K + 3 \rightarrow \text{safe required}$$

$$141 \geq 4K$$

$$\text{hence } K \leq 35$$

So for $K \leq 35$ Ben can avoid the rotten squares.



P.S the last question is probably not TMUA style exactly but I thought it was fun :).